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The effects of short sleep duration on proteinuria and chronic kidney disease: a systematic review and meta-analysis

Wisit Cheungpasitporn[1,*], Charat Thongprayoon[1,*], Maria L. Gonzalez-Suarez[1], Narat Srivali[2], Patompong Ungprasert[3], Wonngarm Kittanamongkolchai[1], Sean M. Caples[2] and Stephen B. Erickson[1]

[1]Division of Nephrology and Hypertension, Mayo Clinic, Rochester, MN 55905, USA, [2]Division of Pulmonary and Critical Care Medicine, Mayo Clinic, Rochester, MN, USA and [3]Department of Rheumatology, Mayo Clinic, Rochester, MN, USA

Correspondence and offprint requests to: Wisit Cheungpasitporn; E-mail: wcheungpasitporn@gmail.com
*These authors contributed equally to this manuscript.

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ORIGINAL ARTICLE

ABSTRACT

Background: The risks of proteinuria and chronic kidney disease (CKD) in adults who regularly have short sleep duration (short sleepers) are controversial. The aim of this meta-analysis was to assess the effects of short sleep duration on proteinuria and CKD.
Methods: A literature search was conducted using MEDLINE, EMBASE and the Cochrane Database of Systematic Reviews from the inception of the databases through November 2015. Studies that reported relative risks, odd ratios or hazard ratios comparing the risks of proteinuria and CKD in short sleepers were included. Pooled risk ratios (RR) and 95% confidence intervals (CI) were computed utilizing a random-effect, generic inverse variance method.
Results: Six observational studies with 252 075 individuals and three observational studies with 37 197 individuals were included in the analyses to assess the risks of CKD and proteinuria in short sleepers, respectively. The pooled RR of CKD in short sleepers was 1.51 (95% CI, 0.99–2.55). When meta-analysis was restricted only to studies with adjusted analysis for confounders assessing the risk of CKD in short sleepers, the pooled RR of CKD was 1.54 (95% CI, 0.80–2.95). The pooled RR of proteinuria in short sleepers was 1.47 (95% CI, 1.26–1.72).
Conclusions: Despite the lack of significant association between short sleep duration and CKD, our meta-analysis suggests a potential association between short sleep duration and proteinuria, a surrogate marker for kidney disease progression. Future study is required to investigate if reversal of short sleep helps reduce proteinuria.

Keywords: chronic kidney disease, proteinuria, short sleep, sleep, sleep duration

INTRODUCTION

Chronic kidney disease (CKD) is an important universal public health problem, and is associated with long-term morbidity and mortality [1–3]. The prevalence of CKD has markedly risen over the past 10 years [4, 5]. In the USA, more than half of all adults over 30 years old are likely to develop CKD in their lifetime, and CKD prevalence is projected to increase from 13.2% currently, to 14.4% in 2020 and to 16.7% in 2030 [6]. Therefore, CKD prevention and management with lifestyle changes and medications continue to be major public health challenges [7–9].

Adequate sleep is crucial for the regulation of body metabolism and various physiological functions. Many people sleep less than 5–6 h a day [10, 11]. Some studies have demonstrated an association between short sleep duration (<7 h) and several comorbid conditions; such as diabetes mellitus, obesity, hypertension and cardiovascular disease [10, 12]. Short sleep duration has been associated with increased mortality [13]. However, the risk of CKD in adults who regularly have short sleep duration (also known as short sleepers) is still conflicting. Several studies have demonstrated an increased risk of CKD in short sleepers [14, 15]. Conversely, few studies have shown no association between short sleep duration and CKD [16–19]. Thus, we conducted this systematic review and meta-analysis to assess the risks of CKD and proteinuria, a surrogate marker for kidney disease progression [20], in short sleepers.

MATERIALS AND METHODS

Search strategy

Two investigators (W.C. and C.T.) independently searched published studies and conference abstracts indexed in MEDLINE, EMBASE and the Cochrane Database from the inception of the databases through November 2015 using the search strategy outlined in Item S1 in Supplementary data. Medical Subject Headings (MeSH) terms were also applied in the search strategy. A manual search for additional pertinent studies using references from retrieved articles was also performed. We followed the Preferred Reporting Items for

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Systematic Reviews and Meta-Analyses (PRISMA) guidelines for a systematic review and meta-analysis [21].

Inclusion criteria

The inclusion criteria were as follows: (i) population/patient: adults (≥18 years old); (ii) intervention/indicator: short sleep; (iii) comparator/control: individuals with adequate sleep; (iv) outcomes: CKD and/or change/appearance of proteinuria; (v) randomized controlled trials (RCTs) or observational studies (case–control, cross-sectional or cohort studies) published as original studies or conference abstracts; and (vi) studies that provided data to calculate odds ratios, relative risks (RR), hazard ratios or standardized incidence ratios with 95% confidence intervals (CIs). No limits were applied to the language.

Study eligibility was independently ascertained by the two investigators noted previously. Differing decisions were solved by mutual consensus. The quality of each study was individually evaluated by each investigator utilizing Newcastle-Ottawa quality assessment scale [22].

Data extraction

We used a standardized data collection form to extract the following information: last name of the first author, study design, year of study, country of origin, year of publication, sample size, patient population, definitions of short sleep duration, CKD, proteinuria and confounder adjustment and adjusted effect estimates with 95% CI.

Statistical analysis

Review Manager 5.3 software (Version 5.3, Copenhagen, Denmark) from the Cochrane Collaboration was used for data analysis. Point estimates and standard errors were obtained from individual studies and were consolidated by the generic inverse variance method of DerSimonian and Laird [23]. Given the high likelihood of inter-study variances, we used a random-effect model rather than a fixed-effect model. Statistical heterogeneity was appraised using the Cochran's Q test. This statistic was complemented with the I^2 statistic, which quantifies the proportion of the cumulative variation across studies that is due to heterogeneity rather than chance. A value of I^2 of 0–25% represents insignificant heterogeneity, 26–50% low heterogeneity, 51–75% moderate heterogeneity and >75% high heterogeneity [24]. The presence of publication bias was assessed by funnel plots of the logarithm of odds ratios versus their standard errors [25].

RESULTS

Our search strategy yielded 9451 articles. Of these, 8974 articles were excluded based on their relevance and eligibility criteria following the review of their title and abstract. The remaining 477 articles underwent full-length article review. From those, 469 articles were excluded (72 articles did not report the outcomes of interest and 397 articles were not observational studies or RCTs). Eight articles were identified and included in data analysis.

Of eight observational studies (two cohort [19, 26] and six cross-sectional studies [14–18, 27]), six studies [14–19] evaluated the risk of CKD in short sleepers and three studies [17, 26, 27] assessed the risk of proteinuria. Five studies performed adjusted analysis for potential risk factors for CKD and/or proteinuria [14, 15, 17, 19, 26]. Item S2 in Supplementary data outlines our search methodology and selection process. Most included studies defined short sleep duration as sleep duration less than 4–5 h/day. Table 1 illustrates the detailed characteristics and quality assessment of the included studies.

Study quality

All observational studies were considered moderate to high quality, with median Newcastle-Ottawa quality assessment scale of 7 (range, 6–9), as shown in Table 1.

The risk of CKD in short sleepers

The majority of the studies included for analysis defined CKD as estimated glomerular filtration rate (eGFR) <60 mL/min/1.73 m^2. The pooled RR of CKD in short sleepers was 1.51 (95% CI, 0.90–2.55, $I^2 = 91\%$). Figure 1 shows the forest plot of the included studies. When meta-analysis was performed excluding two studies, one by Guo *et al.* [15] and the other by Nagai *et al.* [18] (these studies defined short sleep duration as ≤6 and <7.5 h/day, respectively), the pooled RR was 1.70 (95% CI, 0.79–3.65, $I^2 = 94\%$). When meta-analysis was performed excluding a study by Jean-Louis *et al.* [14] (eGFR definition for CKD in this study was unclear), the pooled RR was 1.15 (95% CI, 0.88–1.50). The statistical heterogeneity was low with an I^2 of 42%.

The risk of proteinuria in short sleepers

All included studies defined proteinuria as urinary protein ≥1+ by dipstick or macroalbuminuria. The pooled RR of proteinuria in short sleepers was 1.47 (95% CI, 1.26–1.72). Figure 2 shows the forest plot of the included studies. The statistical heterogeneity was not significant with an I^2 of 0%.

Sensitivity analysis

To minimize the effects of confounders, we performed a sensitivity analysis excluding the studies without adjusted analysis for known risk factors for CKD [16, 18]. The pooled RR of CKD in short sleepers was 1.54 (95% CI, 0.80–2.95, $I^2 = 94\%$).

Evaluation for publication bias

Funnel plot to assess publication bias for the risk of CKD in short sleepers is summarized in Supplementary data, Figure S1. The graph demonstrates no obvious asymmetry and, thus, suggests the absence of publication bias.

DISCUSSION

In this study, despite a trend toward higher CKD risk in short sleepers, our meta-analysis did not find a significant association between short sleep duration and CKD. However, we demonstrated a significant association between short sleep duration and proteinuria, with an overall 1.47-fold increased risk of

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Table 1. Main characteristics of the studies included in this meta-analysis

	Agarwal *et al.* [16]	Yamamoto *et al.* [16]	Fujibayashi *et al.* [17]	Ohkuma *et al.* [17]
Country	USA	Japan	Japan	Japan
Study design	Cross-sectional study	Cohort study	Cross-sectional study	Cross-sectional study
Year	2011	2012	2012	2013
Total number	164	6834	25 493	4870
Study sample	Male veteran aged 18–90 years (19 controls without hypertension, DM or CKD, 145 patients with CKD)	Employees of Osaka University added 20–65 years who visited Osaka University Health Care center for their mandatory annual health examination between April 2006 and March 2010 and did not have eGFR <60 mL/min/1.73 m^2, proteinuria or treatment for self-reported kidney disease	Company employees and their families who participated in annual health checkup	Type 2 diabetic patients aged ≥20 years from multicenter enrolled in Fukuoka Diabetes Registry
Exposure definition	Sleep duration <5 h/night	Sleep duration ≤4 h/night	Sleep duration <5 h/night	Sleep duration <4.5 h/night
Exposure measurement	Actigraphy	Self-administered standardized questionnaires	Self-administered questionnaires. Well-trained staff interviewed any participants who failed to complete the form	Self-administered questionnaires
Outcome definition	CKD; an eGFR of <60 mL/min/1.73 m^2, the presence of proteinuria (spot urine protein creatinine ratio of >200 mg/g creatinine) or the presence of structural kidney disease	Proteinuria: urinary protein ≥1+	CKD; eGFR <60 mL/min/1.73 m^2 Proteinuria urinary protein ≥1+	Albuminuria; UACR ≥30 mg/g Macroalbuminuria; UACR ≥300 mg/g
Outcome ascertainment	Serum creatinine, spot urine analysis	Spot urine dipstick test was interpreted by well-trained nurses	eGFR was calculated using the Japanese GFR inference formula Spot urinalysis was performed using Clinitek Atlas XL autoanalyzer	Urinary creatinine and albumin were measured using an enzymatic method and an immunonephelometry method from spot urine sample
Adjusted OR or RR for	3.83 (0.85–17.28)	1.72 (1.16–2.53)	CKD 1.02 (0.85–1.22) Proteinuria 1.38 (1.15–1.65)	Albuminuria 1.41 (1.02–1.96) Macroalbuminuria 1.83 (1.15–2.91)
Confounder adjusted	None	Yes but not clearly defined	Sex, age, systolic blood pressure, HDL-C, LDL-C, TG, HbA1cm, uric acid, the presence of renal disease, metabolic disorders, arteriosclerotic disease, other unhealthy lifestyle habits (obesity, smoking, irregular meals, infrequent exercise, alcohol intake)	None
Quality assessment (Newcastle-Ottawa scale)	Selection: 3 Comparability: 0 Outcome: 3 Total: 6	Selection: 3 Comparability: 1 Outcome: 3 Total: 7	Selection: 3 Comparability: 2 Outcome: 3 Total: 8	Selection: 4 Comparability: 0 Outcome: 3 Total: 7
	Sasaki *et al.* [19]	Jean-Louis *et al.* [18]	Guo *et al.* [15]	Nagai *et al.* [18]
Country	Japan	USA	China	Japan
Study design	Cohort study	Cross-sectional study	Cross-sectional study	Cross-sectional study
Year	2014	2014	2015	2015
Total number	3600	216 749	5555	514
Study sample	A local healthy government employees aged 35–58 years who underwent annual		Randomly sampled hypertensive patients aged ≥35 years in rural areas of Liaoning	Hypertensive outpatients aged >50 years

Continued

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Table 1. Continued

	Agarwal *et al.* [16]	Yamamoto *et al.* [26]	Fujibayashi *et al.* [17]	Ohkuma *et al.* [27]
	health checkup with eGFR ≥60 mL/min/1.73 m^2	Non-institutionalized US adults from National Health Interview Survey obtained from 1977 through 2009	Province, China between January 2012 and August 2013	
Exposure definition	Sleep duration ≤5 h/night	Sleep duration <5 h/night	Sleep duration ≤6 h/night	Sleep duration <7.5 h/night
Exposure measurement	Self-administered questionnaires	Face-to-face interview	Self-administered questionnaire	Interviewer-administered questionnaire
Outcome definition	Incident CKD; eGFR <60 mL/min/1.73 m^2 after the first checkup	Physician-diagnosed kidney disease	CKD; eGFR <60 mL/min/1.73 m^2	CKD; eGFR <60 mL/min/1.73 m^2
Outcome ascertainment	eGFR was calculated using the Japanese GFR inference formula. Serum creatinine was checked annually	Face-to-face interview	eGFR was calculated using CKD-EPI equation	eGFR was calculated using MDRD equation
Adjusted OR or RR for	0.99 (0.53–1.73)	3.10 (2.38–4.01)	1.70 (1.05–2.73)	0.96 (0.63–1.48)
Confounder adjusted	Age, sex, baseline eGFR, systolic BP, BMI, TC, TG, fasting glucose, uric acid, C-reactive protein, proteinuria, family history of renal disease, smoking, alcohol, exercise, education, work hours and job strain	Age and sex	Age, gender, ethnicity, education level, family income, marital status, diet score, daytime somnolence, smoking, alcohol drinking, physical activity, BMI, waist circumference, systolic BP, diastolic BP, TC, TG, LDL-C, HDL-C, anemia, uric acid, use of ACEI/ARB, history of heart disease and stroke, depressive symptoms and quality of life	None
Quality assessment (Newcastle-Ottawa scale)	Selection: 3 Comparability: 2 Outcome: 3 Total: 8	Selection: 4 Comparability: 1 Outcome: 2 Total: 7	Selection: 4 Comparability: 2 Outcome: 3 Total: 9	Selection: 3 Comparability: 0 Outcome: 3 Total 6

ACEI/ARB, angiotensin-converting enzyme inhibitors/angiotensin-receptor blockers; BMI, body mass index; BP, blood pressure; Ca, calcium; DM, diabetes mellitus; HAM-D, The Hamilton Depression Rating Scale; HbA1c, glycated hemoglobin; HDL-C, high-density lipoprotein cholesterol; LDL-C, low-density lipoprotein cholesterol; Mg, magnesium; n/a, not available; OR, odds ratio; TC, total cholesterol; TG, triglyceride; UACR, urine albumin to creatinine ratio.

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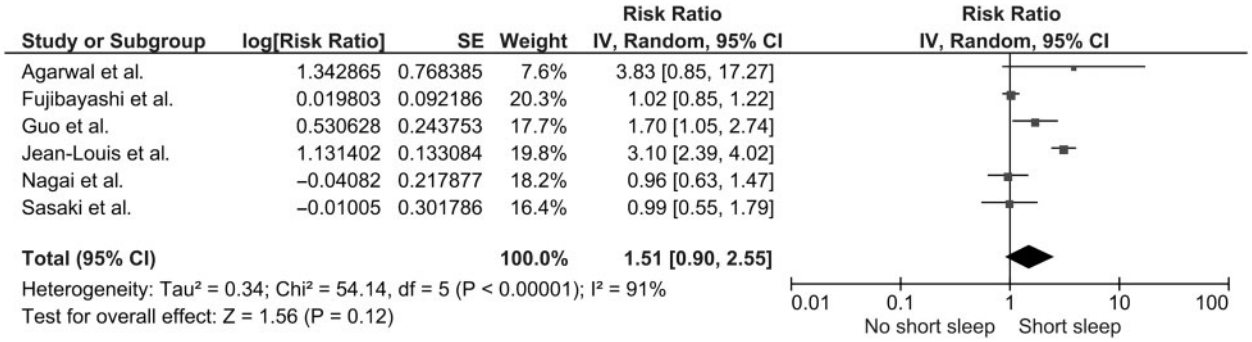

FIGURE 1: Forest plot of the all included studies comparing risk of CKD in short sleepers and non-short sleepers; square data markers represent RR; horizontal lines, the 95% CIs with marker size reflecting the statistical weight of the study using random-effects meta-analysis. A diamond data marker represents the overall RR and 95% CI for the outcome of interest. IV, inverse variance; SE, standard error.

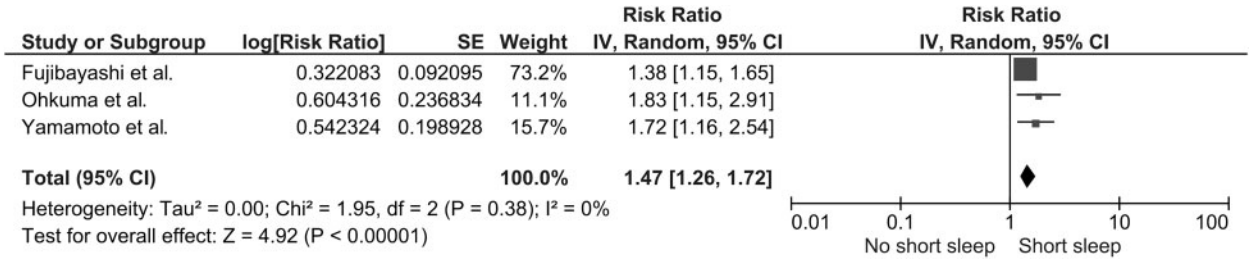

FIGURE 2: Forest plot of the all included studies comparing risk of proteinuria in short sleepers and non-short sleepers; square data markers represent RR; horizontal lines, the 95% CIs with marker size reflecting the statistical weight of the study using random-effects meta-analysis. A diamond data marker represents the overall RR and 95% CI for the outcome of interest. IV, inverse variance; SE, standard error.

proteinuria. Most of the included studies defined the significant short sleep duration as approximately <5 h per day.

Circadian timing system is critically implicated in the maintenance of physiologic functions related to the kidney, including blood pressure and GFR [28, 29]. Sleep is a key regulator of these physiological functions [5]. During normal sleep, reduced sympathetic activity and increased vagal tone, particularly during non-rapid eye movement sleep, are responsible for the nocturnal dipping of blood pressure associated with sleep. Thus, reduced sleep duration may lead to sympathetic nervous system stimulation and attenuation of the sleep-induced decrease in blood pressure [30, 31]. Recent studies have shown higher risk of hypertension in individuals with short sleep duration [32–34]. Sleep also has an important impact on the release of metabolic hormones, and their responsiveness. [15]. Short sleep has also been shown to be associated with obesity and diabetes [12]. Despite the fact that many of the individuals included in this meta-analysis had hypertension and/or diabetes, two known common causes of CKD, our study did not find a significant association between short sleep duration and CKD, defined by eGFR <60 mL/min/1.73 m^2. It is possible that the negative impact of reduced sleep duration itself on kidney function is small.

Proteinuria is considered as a strong predictor of clinical progression of CKD. Proteinuria in patients with CKD may accelerate kidney disease progression to end-stage kidney disease [35–37]. Although there was not a significant association between short sleep duration and CKD, our meta-analysis demonstrated a significantly increased proteinuria in short sleepers. Future study is needed to assess if short sleep duration worsens kidney function in CKD patients.

Although almost all included studies were of moderate to high quality, there are some limitations. First, both short and long sleep have been shown to be associated with mortality [13]. Due to limited data, we only investigated the effect of short sleep on CKD. Thus, future studies on the impact of long sleep duration on kidney function are required. Second, obstructive sleep apnea (OSA) has now been considered as an independent risk factor for CKD [5]. Although body mass index and obesity were included in the adjusted analysis of included studies, OSA and quality of sleep were still important potential confounders. Thus, future studies on the effect of quality of sleep on kidney function are needed. Third, Sasaki *et al.* [19] recently reported that short sleep duration was associated with a significantly higher risk of CKD among shift workers. It is possible that short sleep duration raises CKD risk in this specific patient population. Fourth, although this meta-analysis of observational studies with confounder adjusted analysis helps remove potential bias, the lack of RCTs is an important limitation (all included studies are observational studies), and a causal relationship cannot be inferred. Lastly, there is statistical heterogeneity in the analysis of CKD risk in short sleepers. The difference in the definition of CKD and short sleep duration may be the main sources of this heterogeneity. However, in the meta-analysis excluding the study by Jean-Louis *et al.* [14], the heterogeneity was low, and the risk of CKD in short sleepers was still not significant. Besides, there was no significant heterogeneity in the meta-analysis of the risk of proteinuria in short sleepers.

In conclusion, despite the lack of association between short sleep duration and CKD, our meta-analysis suggests a potential

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increased proteinuria in short sleepers. This finding may impact the clinical management of CKD.

SUPPLEMENTARY DATA

Supplementary data are available online at http://ndt.oxfordjournals.org.

AUTHORS' CONTRIBUTIONS

All authors had access to the data and a role in writing the manuscript.

CONFLICT OF INTEREST STATEMENT

We do not have any financial or non-financial potential conflicts of interest.

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